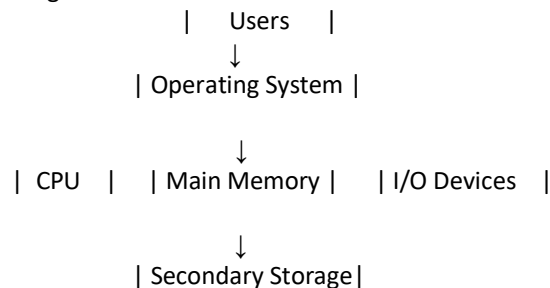


UNIT – I Long Answers

1. Explain Computer-System Organization with Diagram

Definition: Computer System Organization refers to the arrangement and interaction of hardware components such as CPU, memory, I/O devices, and storage devices under the control of an Operating System.

Diagram:



Components

1. Processor (CPU):

Executes instructions.

Controls system operations.

Performs arithmetic and logical operations.

2. Main Memory:

Stores programs and data currently being executed.

Provides fast access to CPU.

3. I/O Devices

Input devices: Keyboard, Mouse, Scanner.

Output devices: Monitor, Printer.

4. Secondary Storage

Hard Disk, SSD, Optical Disk.

Stores data permanently.

5. Operating System:

Manages all hardware and software resources.

Acts as an interface between user and hardware.

Advantages:

Efficient resource utilization.

Better performance.

Easier management of devices.

2. Explain Batch Processing System

Definition: A Batch Processing System executes a group of jobs together without user interaction.

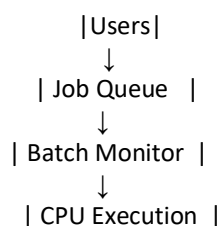
Working:

Users submit jobs.

Jobs are collected into batches.

The operating system executes jobs one after another.

Diagram:



Advantages:

- High throughput.
- Reduced setup time.
- Suitable for large repetitive jobs.

Disadvantages:

No user interaction.

Long turnaround time.

Difficult debugging.

Applications:

Payroll processing.

Bank statement generation.

Examination result processing.

3. Write a Note on Multiprogrammed Systems

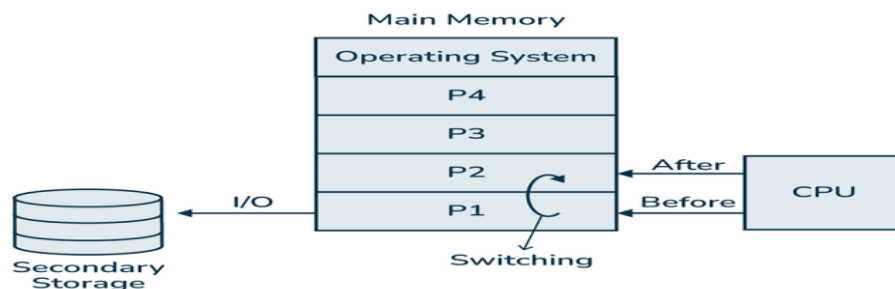
Definition: Multiprogramming is a technique in which several programs are kept in memory simultaneously and the CPU executes them alternately.

Working:

When one process waits for I/O, CPU executes another process.

CPU utilization increases.

Diagram



Advantages:

Better CPU utilization.

Increased throughput.

Reduced idle time.

Disadvantages:

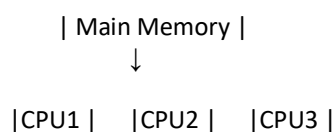
Complex scheduling.

Requires memory management.

4. Explain Multiprocessor System with Diagram

Definition: A Multiprocessor System contains two or more processors sharing memory and other resources.

Diagram:



Types:

Symmetric Multiprocessing (SMP):

All processors are equal.

Each processor performs the same functions.

Asymmetric Multiprocessing (AMP):
One master processor controls others.

Advantages:
Increased reliability.
Higher throughput.
Faster execution.

Disadvantages
Complex OS design
Expensive hardware.

5(a). Real Time System

Definition: A Real-Time Operating System (RTOS) guarantees response within a specified time limit.

Types

Hard Real-Time System
Missing deadlines is unacceptable.
Example: Air Traffic Control.

Soft Real-Time System
Missing deadlines occasionally is acceptable.
Example: Multimedia systems.

Advantages:
Fast response.
Predictable behavior.

Applications:
Medical equipment.
Industrial control systems.

5(b). Time Sharing System

Definition: A Time Sharing System allows many users to share the CPU simultaneously.

Working:
CPU time is divided into small time slices.
Each user gets CPU time periodically.

Advantages:
Quick response.
Efficient resource sharing.

Disadvantages:
Security issues.
High overhead.

Examples:
UNIX
Linux

6. Functions of Components of Operating System

a) Process Management

Functions:
Process creation and deletion.
CPU scheduling.
Synchronization.

Deadlock handling.

b) Memory Management

Functions:

Allocation and deallocation.

Tracking memory usage.

Paging and segmentation.

c) File Management

Functions:

File creation.

File deletion.

Directory management.

Access control.

d) Secondary Storage Management

Functions:

Free space management.

Storage allocation.

Disk scheduling.

e) Networking Management

Functions:

Communication between systems.

Resource sharing.

Network security.

f) I/O System Management

Functions:

Device management.

Buffering.

Spooling.

Caching.

7. Explain Any Five Operating System Services

1. Program Execution

Loads and executes programs.

2. I/O Operations

Controls communication with devices.

3. File System Manipulation

Creates, deletes, and manages files.

4. Communication

Allows data exchange between processes.

5. Error Detection

Detects and handles errors.

Additional Services:

Resource Allocation

Accounting

Protection and Security

8. Explain Any Five File Operations

1. Create

Creates a new file.

2. Open

Makes a file available for use.

3. Read

Retrieves data from a file.

4. Write

Stores data into a file.

5. Delete

Removes a file permanently.

(Explain in your words)

9. Explain Various File Attributes

1. Name

Identifies the file.

2. Identifier

Unique file number.

3. Type

Specifies file category.

4. Location

Address of file storage.

5. Size

File length in bytes.

6. Protection

Access permissions.

7. Time and Date

Creation and modification details.

8. Owner

User who created the file.

10. Explain Common File Types, Extensions and Functions

File Type	Extension	Function
Text File	.txt	Stores plain text
Document	.doc/.docx	Word processing
PDF	.pdf	Portable document
Image	.jpg/.png	Stores pictures
Audio	.mp3	Sound files
Video	.mp4	Multimedia files

11. Explain Various Access Methods in Files

1. Sequential Access

Records accessed in order.

Record1 → Record2 → Record3

Advantage: Simple implementation.

2. Direct (Random) Access

Any record can be accessed directly.

Record 5

Record 10

Record 20

Advantages: Faster access.

3. Indexed Access

Index table points to records.

Advantages: Efficient searching.

4. Hashed Access

Hash function determines record location.

Advantages: Very fast retrieval.

12. Explain Any Five Directory Operations

1. Create Directory
Creates a new folder.
2. Delete Directory
Removes a folder.
3. Search Directory
Locates files.
4. Rename Directory
Changes directory name.
5. List Directory Contents
Displays files and subdirectories.

13. Explain Different Types of Access Control for Files

1. Read Access
Allows reading data.
 2. Write Access
Allows modification.
 3. Execute Access
Allows program execution.
 4. Append Access
Allows adding data.
 5. Delete Access
Allows file removal.
- Access Control Methods:
Access Control List (ACL)
Password Protection
User/Group Permissions

14. Explain Allocation Methods

A. Contiguous Allocation

Definition: Files occupy consecutive disk blocks.

File A

[10][11][12][13]

Advantages

Fast access.

Simple implementation.

Disadvantages

External fragmentation.

Difficult file expansion.

B. Linked Allocation

Definition

File blocks are linked together.

10 → 25 → 7 → 40

Advantages:

No external fragmentation.

Easy file growth.

Disadvantages:

Slow random access.

Pointer overhead.

C. Indexed Allocation

Definition: An index block stores addresses of all file blocks.

Index Block

|--> 10

|--> 25

|--> 40

Advantages:

Supports direct access.

No external fragmentation.

Disadvantages

Requires extra index block.

15. Explain Any Two Free-Space Management Methods

A. Bit Vector Method

Definition:

Each disk block is represented by a bit.

1 1 0 0 1 0 1

1 → Free

0 → Occupied

Advantages:

Easy implementation.

Fast search.

Disadvantages:

Requires extra memory.

B. Linked List Method

Definition

All free blocks are linked together.

5 → 12 → 20 → 25 → NULL

Advantages:

No extra bitmap required.

Disadvantages: Slow traversal.

C. Grouping Method

Definition:

A free block contains addresses of several other free blocks.

Advantages:

Faster allocation.

D. Counting Method

Definition:

Stores starting block and number of consecutive free blocks.

(Start Block = 100, Count = 20)

Advantages:

Efficient for contiguous free spaces.

Less storage overhead.